
Few-Step Consistency Distillation for Crystal Structure Diffusion Models

Extended Abstract Track Submissions

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Abstract

Consistency distillation collapses many-step diffusion sampling into a handful of direct network calls by training a student to map any noised point on a trajectory straight to its clean endpoint. Applying this to DiffCSP, a joint diffusion model over a crystal’s lattice (Euclidean, variance-preserving) and fractional atomic coordinates (periodic, variance-exploding), requires a boundary-satisfying consistency function for each track, and multistep consistency sampling (Song et al. 2023, Algorithm 1) exercises those functions in a way that ordinary diffusion sampling and consistency-distillation training never do: by renoising the network’s own previous estimate and calling it again. We show that the direct DDIM/Tweedie lattice estimate — exact at the noise-free boundary, and a natural first derivation because no separate reparameterization initially appears necessary — has an error-amplification factor that is unbounded as noise grows, and confirm this mechanistically at the schedule’s practical maximum-noise point on the pretrained teacher, with no additional training required to see it. A bounded, EDM-style $c_{\text{skip}}/c_{\text{out}}$ reparameterization removes this specific defect; however, replaying the same noise draws through both estimators shows the multistep trajectory largely self-corrects a large first-step error at subsequent, less severe steps, so the reparameterization does not change the sampler’s end-to-end validity. It does not need to: the coordinate track has a second, independent failure that alone invalidates every sample at every step budget we tried, on either lattice estimator. Its estimator’s coefficients are already bounded — ruling out the same fix — and the network instead extrapolates poorly on the self-referential, renoised inputs multistep sampling produces, inputs its training distribution, built only from real data, never included; we report this as an open problem after ruling out the obvious stopgap (clamping the network’s raw output) as degenerate. Together these results locate two distinct, previously unremarked failure modes in adapting consistency distillation to a coupled non-Euclidean/periodic diffusion process, mechanistically diagnose both, resolve one in isolation while showing why that resolution does not by itself fix end-to-end sampling, and leave the other as a precisely characterized open problem — alongside a confirmed reproduction of the live-student-versus-EMA-target instability the original Consistency Models paper reports for single-track image diffusion.

1 Introduction

Crystal structure prediction (CSP) — given a chemical composition, predicting the periodic atomic arrangement that minimizes free energy — is a central task in computational materials discovery. DiffCSP [1] formulates CSP as a joint equivariant diffusion process over a crystal’s lattice and fractional atomic coordinates, using a periodic-boundary graph neural network (CSPNet) as its denoiser. Like most diffusion samplers, DiffCSP’s default sampling procedure uses 1,000 sequential denoising steps, each requiring a full network forward pass. This cost is a practical bottleneck for high-throughput screening, where thousands to millions of candidate structures must be sampled.

Consistency distillation [2] trains a student network to map any point on a diffusion trajectory directly to its clean endpoint, collapsing many-step sampling into a handful of steps. It has been applied successfully to image diffusion models, whose forward processes are single-track Gaussian corruptions. DiffCSP’s forward process is not: it couples two structurally different noise processes — a variance-preserving (VP) process on the lattice matrix and a variance-exploding (VE) process on fractional coordinates constrained to a periodic domain — with no existing distillation recipe for either the individual tracks or their combination.

We set out to train and evaluate a consistency-distilled student for both tracks. In building the sampler needed to actually exercise a consistency function — Algorithm 1 of Song et al. [2]: one direct network call per step, producing a boundary-condition estimate, with that estimate renoised before the next call, as opposed to the many small local steps an ordinary diffusion solver takes — we found that neither track’s boundary-satisfying estimator, as we had first derived or DiffCSP’s own code already computes it, actually supports this without producing geometrically invalid structures, regardless of how well the network is trained: an *untrained* teacher network fails identically to a fine-tuned student under the same estimator. Tracing why is this paper’s main content. The two tracks fail for different reasons, one of which we resolve without any additional training and one we do not. We also confirm, in the course of building the training procedure itself, a specific instability Song et al. [2] already document for their own single-track setting, carried over unresolved into an early version of our training code.

2 Method

Background. DiffCSP’s denoiser D_θ takes a noised lattice $\ell_t \in \mathbb{R}^{3 \times 3}$, noised fractional coordinates $x_t \in [0, 1)^{3N}$ for N atoms, atom types, and a timestep t , and returns $(\hat{\epsilon}_\ell, \hat{s}_x)$: a noise-prediction for the lattice and a normalized score for the coordinates. The lattice follows a VP forward process, $\ell_t = \sqrt{\bar{\alpha}_t} \ell_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, with $\bar{\alpha}_t$ from a cosine schedule. The coordinates follow a VE forward process on the torus, $x_t = (x_0 + \sigma_t z) \bmod 1$, denoised at inference via an annealed-Langevin predictor step.

A first derivation, and why it looks sufficient. DiffCSP’s own zero-padded schedules give $\bar{\alpha}_0 = 1$ and $\sigma_0 = 0$ exactly. The direct DDIM/Tweedie estimate for the lattice,

$$\hat{\ell}_0 = \frac{\ell_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}_\ell}{\sqrt{\bar{\alpha}_t}},$$

satisfies $\hat{\ell}_0(\ell_0, \bar{\alpha}_t=1) = \ell_0$ exactly, for any $\hat{\epsilon}_\ell$: the boundary condition a consistency function requires. The analogous Tweedie estimate for the coordinates, $\hat{x}_0 = (x_t + \sigma_t^2 \hat{s}_x) \bmod 1$, satisfies its own boundary condition ($\sigma_0 = 0$) the same way. Both hold without introducing the separate $c_{\text{skip}}, c_{\text{out}}$ reparameterization Song et al. [2] use (following Karras et al. [3]); each track’s existing schedule already supplies a valid consistency function by this construction, so skipping that reparameterization looks like a reasonable simplification rather than an omission. It is not: Sec. 3 shows this direct estimate is, in fact, exactly the failure mode $c_{\text{skip}}/c_{\text{out}}$ exists to prevent.

Training. Following Algorithm 2 of Song et al. [2]: for a clean batch (ℓ_0, x_0) , sample an adjacent index pair $t_n < t_{n+1}$ on a uniform N -point grid (excluding the terminal index, where the cosine schedule’s clipped $\bar{\alpha}$ destabilizes the DDIM estimate), forward-noise both tracks to t_{n+1} , and use the frozen teacher for one deterministic step down to t_n . The student’s consistency function at $(\ell_{t_{n+1}}, x_{t_{n+1}}, t_{n+1})$ is regressed, by summed per-track MSE, against the same function evaluated by a separate target network at the teacher-denoised $(\hat{\ell}_{t_n}, \hat{x}_{t_n}, t_n)$.

Target-network design is not incidental. Letting the target network *be* the live student, our first implementation, trained stably on a small pilot but diverged smoothly at full dataset scale (27,136 structures, 15 epochs: loss $1,249 \rightarrow 2.9 \times 10^5$) even with gradient clipping and $\text{lr} = 10^{-6}$ — clipping bounds an update’s magnitude, not its systematic directional bias, which a self-referential target compounds over many steps too gradually for a small pilot to reveal. An EMA target ($\theta^- \leftarrow \mu \theta^- + (1 - \mu) \theta$, $\mu = 0.999$) eliminates this entirely. Song et al. [2] report the identical $\theta^- = \theta$ instability and fix for their own single-track image setting; we confirm it reproduces for a dual-track, graph-structured model, not a new instability.

Multistep consistency sampling exposes what training does not. Algorithm 1 of Song et al. [2] samples by direct jump-and-renoise: given noise (ℓ_T, x_T) , one network call produces boundary-condition estimates $(\hat{\ell}_0, \hat{x}_0)$ at the maximum usable noise index; each subsequent step renoises $(\hat{\ell}_0, \hat{x}_0)$ to the next grid index and calls the network again. This differs fundamentally from resampling the deterministic DDIM/probability-flow-ODE solver onto a coarser sub-grid (a valid, separate step-compression technique that does not exercise a consistency function’s boundary-condition property at all, since it never renoises its own output): the solver takes small local steps between adjacent, still-plausible states, while Algorithm 1 makes a single global jump to a full denoising estimate and then *renoisesthat estimate itself* — a self-referential loop that training, which regresses the consistency function only against genuinely noised *real* data, never rehearses. This distinction is invisible from the training loss alone, and only surfaces once Algorithm 1 is actually run.

3 Two Failures, Not One

Running Algorithm 1 against the frozen teacher — with no consistency-distillation training involved at all — already produces structures with atoms overlapping to within a small fraction of an Ångström, at every step budget we tried. A consistency-distilled student we trained separately (Sec. 2) performs statistically indistinguishably from this untrained teacher at the same one-shot denoising task, across the full noise range, on real held-out structures (e.g. a representative comparison at $\bar{\alpha}_{999}$: teacher mean absolute lattice error 14.9 versus student 14.4; at $\bar{\alpha}_{100}$: teacher coordinate error 0.80 versus student 0.70) — ruling out a training-specific defect and pointing at the sampler or the boundary formulas themselves, which we can therefore diagnose entirely on the frozen, pretrained network. We isolated two independent causes, one in each track, both diagnosable and fixable (or not) without training a student at all.

Lattice: a confirmed, mechanistic error-amplification factor that a multistep trajectory partially self-corrects. The direct DDIM/Tweedie estimate’s $\hat{\epsilon}_\ell$ -coefficient, $\sqrt{1 - \bar{\alpha}_t}/\sqrt{\bar{\alpha}_t}$, is exact at the boundary but *unbounded* as $\bar{\alpha}_t \rightarrow 0$: at DiffCSP’s own $\bar{\alpha}_{999} = 2.428 \times 10^{-6}$ (its practical maximum-noise index), this coefficient is ≈ 642 . A prediction error of realistic magnitude for a trained $\hat{\epsilon}_\ell$ head is amplified past any physically plausible lattice scale at this specific step. We confirmed this is a property of the *estimator* at that step, not of network quality: feeding the *teacher* genuine forward-noised ground truth at $\bar{\alpha}_{999}$ and regressing $\hat{\ell}_0$ against it with an aggressive learning rate for 200 steps, loss oscillates ($323 \rightarrow 137 \rightarrow 230 \rightarrow \dots \rightarrow 237$) rather than decreasing, and the post-training error is unchanged from the untrained baseline. We replace the direct estimate with a bounded, $c_{\text{skip}}/c_{\text{out}}$ reparameterization in the style of Karras et al. [3] and Song et al. [2], adapted to the lattice’s VP schedule via the standard VP $\leftrightarrow$ VE noise-level identity $\sigma_{\text{eff}}^2 = (1 - \bar{\alpha}_t)/\bar{\alpha}_t$:

$$c_{\text{skip}}(\bar{\alpha}_t) = \frac{\sigma_{\text{data}}^2 \sqrt{\bar{\alpha}_t}}{(1 - \bar{\alpha}_t) + \sigma_{\text{data}}^2 \bar{\alpha}_t}, \quad c_{\text{out}}(\bar{\alpha}_t) = \frac{-\sigma_{\text{data}} \sqrt{1 - \bar{\alpha}_t}}{\sqrt{(1 - \bar{\alpha}_t) + \sigma_{\text{data}}^2 \bar{\alpha}_t}},$$

with $\sigma_{\text{data}} \approx 3.33$ measured directly as the empirical standard deviation of real MP-20 lattice-matrix entries, preserving the exact boundary condition ($c_{\text{skip}}(1)=1$, $c_{\text{out}}(1)=0$) while $c_{\text{out}}(\bar{\alpha}_t) \rightarrow -\sigma_{\text{data}}$, a finite constant, as $\bar{\alpha}_t \rightarrow 0$, rather than diverging. At the first Algorithm 1 step in isolation this is a large, measurable improvement (a representative single structure’s $\hat{\ell}_0$ magnitude drops from 29.8 to 7.4 at $\bar{\alpha}_{999}$, for the same network output). Over the *full* 16-step trajectory, however, this improvement washes out almost completely: the renoise-and-rejump loop forward-noises whichever $\hat{\ell}_0$ came out of the previous step, and every subsequent grid point has a far smaller amplification factor than the first ($\approx 9.4\times$ at the second point, versus $642\times$ at the first) — concretely, feeding both estimators the same noise draws and the same teacher, the two trajectories’ per-step lattice magnitudes start 43% apart at step four, fall below a 5% relative gap by step eleven, and across 30 real structures the bounded estimate’s final mean absolute error against ground truth (10.14) is not distinguishably better than the direct estimate’s (10.49). The mechanism we identify is real, and we verified it as an isolated, mechanistic effect on the very first step; we did not find that resolving it, on its own, changes the trajectory’s final output enough to matter, because a second, independent, larger failure (below) already invalidates every sample at every step budget we tried, on either lattice estimator.

Coordinates: an open problem, not a coefficient blow-up. The coordinate estimate’s own coefficient, σ_t^2 (further scaled by the network’s un-normalization factor $\sqrt{\sigma_{\text{norm},t}}$), stays under 0.2 across

Table 1: Fraction of 30 real MP-20 structures producing a geometrically valid sample under Algorithm 1 at 16 steps, pretrained teacher only, one seed. “Direct” is the unbounded DDIM/Tweedie lattice estimate; “Bounded” is our $c_{\text{skip}}/c_{\text{out}}$ reparameterization (Sec. 3). “+ true coords” substitutes ground-truth fractional coordinates for the network’s own (independently failing) coordinate estimate, isolating the lattice track; “+ est. coords” uses the network’s coordinate estimate as Algorithm 1 actually produces it end to end. Mean lattice error is the mean, across all 30 structures, of the final step’s elementwise maximum absolute deviation from the real ground-truth lattice matrix.

Configuration	Valid	Mean lattice error
Direct lattice estimate + true coords	30/30	10.49
Bounded lattice estimate + true coords	30/30	10.14
Direct lattice estimate + est. coords (end to end)	0/30	—
Bounded lattice estimate + est. coords (end to end)	0/30	—

the entire sampling grid: there is no analogous blow-up to bound, and a $c_{\text{skip}}/c_{\text{out}}$ reparameterization here changes nothing, since $c_{\text{skip}}=1$ and $c_{\text{out}}=\sigma_t^2$ already is such a blend, with a coefficient that was never the problem. Instead, the raw network output grows far outside its own real range over the course of an Algorithm 1 trajectory: on real forward-noised data, the network’s score output ranges from $\approx 10^{-3}$ near maximum noise to ≈ 475 near zero noise, a smooth, well-behaved function of t ; under Algorithm 1’s renoise-and-rejump loop, the same network’s output reaches ≈ 929 at a step where the in-distribution scale is a small fraction of that. The training objective regresses the consistency function only against real, forward-noised ground truth; it never presents the network with a renoised *estimate* of its own, so nothing in training corrects this extrapolation failure. We considered and rejected a fixed clamp on the raw score as a stopgap: because the in-distribution scale varies by roughly three orders of magnitude across the trajectory, any single clamp value either does nothing early or discards the network’s actual output entirely later — we measured 100% of score entries clamped at every step past the second under a clamp value chosen to look reasonable on the trajectory’s first two steps alone, which produces geometrically valid but essentially undenoised structures, indistinguishable in composition from the initial noise. We report this failure, its cause, and this ruled-out fix, without a resolution: the tractable next step is exposing the network to renoised self-estimates during training, not a change to the sampling-time formula, and we leave verifying that to future work.

4 Experiments

All results below use only the frozen, official pretrained DiffCSP CSP-task checkpoint [1] under Algorithm 1; every checkpoint our training procedure (Sec. 2) has produced is downstream of the still-open coordinate failure in exactly the way Algorithm 1 sampling itself is, so we withhold its match rate to avoid misattributing a sampling-time failure to distillation. Structures use MP-20 real ground-truth compositions [4, 5], via a seeded random sample of the test split (not the first n rows in file order, which differs measurably in composition from the rest). Validity (pymatgen): a structure constructs (finite, positive-volume lattice) and every interatomic distance exceeds $0.5 \text{ \AA}$.

The isolated lattice-only result and the end-to-end result disagree, and both are real. With true coordinates, both estimators reach 30/30: pymatgen’s length/angle reconstruction is insensitive to a lattice matrix’s determinant sign (lengths are row norms; angles come from an $\arccos$ of a ratio), so the 20/30 negative-determinant lattices we measure under *both* estimators do not fail this validity check on their own, and their mean errors against ground truth (10.49 versus 10.14) are not distinguishably different at this sample size — consistent with Sec. 3’s finding that the trajectory dilutes a large first-step error faster than the amplification factor alone predicts. With the network’s own coordinate estimate, both variants instead produce 0/30 valid structures, entirely on the interatomic-distance check. We report both numbers, and not only the more flattering isolated one, deliberately: we return to no match rate against ground truth in this submission, since Algorithm 1 does not produce valid structures end to end under either estimator, and a match rate would measure only how the coordinate failure interacts with pymatgen’s structure matcher on this particular checkpoint — not a property attributable to anything derived here.

5 Related Work

DiffCSP [1] and CDVAE [4] established diffusion and score-based generative models for CSP; we build directly on the former’s released checkpoint and evaluation protocol. Consistency models [2], built on the $c_{\text{skip}}/c_{\text{out}}$ preconditioning of Karras et al. [3], and latent consistency models [6] introduced few-step distillation for image diffusion; all three assume a single Gaussian forward process on an unconstrained Euclidean space. We extend the underlying distillation objective to a coupled VP/VE, Euclidean/periodic setting, and show that the bounded preconditioning Karras et al. [3] and Song et al. [2] use for exactly this kind of numerical reason is not optional convention but a necessary component once the target space is periodic and the noise schedule’s boundary is approached during sampling rather than only during training. To our knowledge, no prior work applies consistency distillation to a crystal-structure or materials-generation diffusion model, and no prior work isolates a failure of this kind (network-independent and sampler-induced, diagnosable on a frozen pretrained network before any distillation training) in a non-Euclidean or periodic diffusion track specifically.

6 Conclusion

We derive and validate consistency-distillation training for DiffCSP’s dual-track, mixed-geometry denoising process, and confirm that the live-student-as-target instability Song et al. [2] report for single-track image diffusion reproduces in this graph-structured, dual-track setting. Attempting to actually exercise a trained consistency function via multistep sampling (Algorithm 1) surfaces two further, independent failures invisible to the training loss, on the frozen teacher alone. The lattice track’s direct boundary estimate has an unbounded error-amplification factor that we trace to its exact mechanism and confirm at the schedule’s practical maximum-noise point; a bounded reparameterization removes this specific defect, but replaying identical noise draws through both estimators shows the multistep trajectory dilutes the error faster than the amplification factor alone would suggest, so the fix does not change end-to-end validity, because the coordinate track’s independent failure already determines it regardless. That second failure has the same symptom but a different, still-open cause — self-referential extrapolation on renoised inputs never seen in training — which we characterize precisely and report without a fix, having ruled out the simplest candidate (a fixed output clamp) as degenerate rather than claiming success it does not have. We see the diagnostic method as the more durable contribution of the two specific numbers: isolating a sampler-level failure on a frozen, pretrained network, before spending any training compute, distinguishes an estimator problem from a training problem cheaply and is directly reusable by any dual-track or non-Euclidean adaptation of consistency distillation, whether or not the specific amplification factor or clamp fraction here transfer. Future work includes exposing the network to renoised self-estimates during training as a candidate fix for the coordinate track, and, once that is resolved, the match-rate evaluation this submission deliberately withholds.

Code and the exact commands used to produce every number in this paper are available at [ANONYMIZED CODE LINK – anonymous.4open.science mirror per the LoG double-blind policy; deanonymize to github.com/DIVkov575/diffcsp-consistency-distillation on acceptance].

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